Linux Troubleshooting Lab

1. Project Overview

Objective: Practice basic system diagnostics and document troubleshooting procedures on Linux Mint 22.3 Cinnamon.
Environment: Lenovo ThinkCentre M710s — Linux Mint 22.3 Cinnamon

2. Test Record

Test 01 — System Identification

Operating system: Linux Mint 22.3 Cinnamon
Command used: *hostnamectl*
Expected observation: The command displays the computer's hostname, operating system, kernel version, hardware architecture, and hardware manufacturer/model where available.Actual observation: The terminal identified the hostname as *ThinkCentre-M-710s*, operating system as *Linux Mint 22.3*, kernel as *7.0.0-38-generic*, architecture as *x86-64*, and hardware model as *ThinkCentre M710s*.Result: Pass

Test 02 — Network Loopback

Operating system: Linux Mint 22.3 Cinnamon
Command used: ping –c 4 127.0.0.1
Expected observation: The local loopback address responds to all four ICMP requests without packet loss
Actual observation: Four packets were transmitted, four were received, and packet loss was 0%. The reported average round-trip time was *0.062 ms*.
Result: Pass

Test 03 — Disk Space

Operating system: Linux Mint 22.3 Cinnamon
Command used: df -h
Expected observation: The command displays filesystem capacity, used space, available space, usage percentage, and mount points, in human-readable points.
Actual observation: The root filesystem */dev/mapper/vgmint-root* reported *108 GB* total, *57 GB* used, *46 GB* available, and *56%* usage. The */boot/efi* filesystem showed *511 MB* total, *6.2 MB* used, and *2%* usage.
Result: Pass

Test 04 — Memory Usage

Operating system: Linux Mint 22.3 Cinnamon
Command used: *free -h*
Expected observation: The command displays memory and swap statistics in human-readable units.
Actual observation: The output reported approximately *19 GiB* total RAM, *4.0 GiB* used, *12 GiB* free, *3.2 GiB* in the buff/cache column, and *15 GiB* available. Swap totalled *1.9 GiB*, w/ *0 B* used.
Result: Pass

Test 05 — Network Configuration

Operating system: Linux Mint 22.3 Cinnamon

Command used: *ip addr*
Expected observation: The command displays network interfaces, interface states, and configured IP addresses.
Actual observation: The output showed the *lo* loopback interface, the *enp1s0* Ethernet interface marked UP, and a *docker0* interface marked DOWN. IP address information was displayed for the listed interfaces w/ sensitive values redacted in the screenshot.
Result: Pass

3. Findings

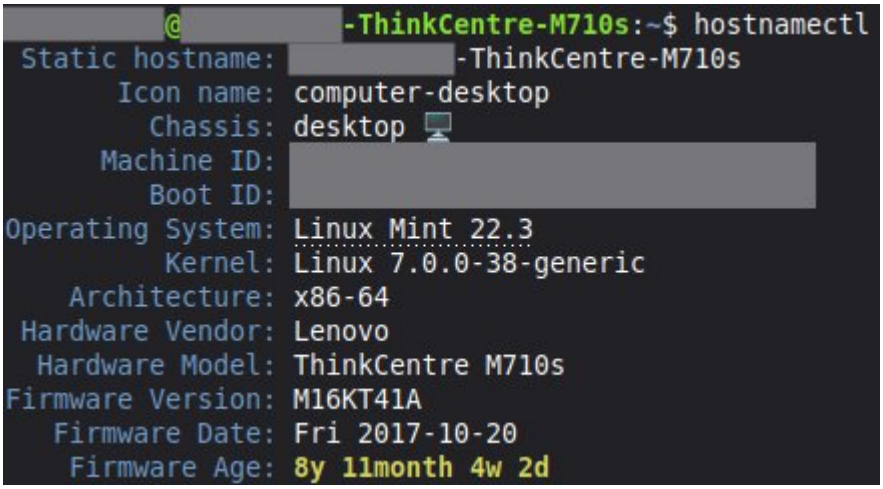
The initial **Linux Mint 22.3** diagnostic tests successfully retrieved system identification, network interface information, memory statistics, and filesystem usage. The loopback test transmitted and received **four packets with 0% packet loss**. The root filesystem reported **56% usage**, with approximately **46 GB available**. Memory information showed approximately **15 GiB available** and **no swap currently in use**. These results establish a baseline for further troubleshooting; they do not independently confirm external internet connectivity or the absence of hardware problems.

4. Lessons Learned

I practiced using Linux terminal commands to identify system hardware and software, test local network communication, inspect network interfaces, review RAM and swap usage, and assess filesystem capacity. I learned that a successful loopback test does not establish external network connectivity, that Linux memory caching affects memory statistics, and that disk usage must be interpreted alongside the available space and filesystem mount point.

5. Evidence

System Information



Network Loopback

```
[redacted]@[redacted]-ThinkCentre-M710s:~$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.061 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.054 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3070ms
rtt min/avg/max/mdev = 0.054/0.062/0.073/0.006 ms
```

Disk Space

```
[redacted]@[redacted]-ThinkCentre-M710s:~$ df -h
Filesystem                Size  Used Avail Use% Mounted on
tmpfs                     2.0G  1.8M  2.0G   1% /run
efivarfs                  128K   60K   64K  49% /sys/firmware/efi/efivars
/dev/mapper/vgmint-root   108G   57G   46G  56% /
tmpfs                     9.8G   24K  9.8G   1% /dev/shm
tmpfs                     5.0M   12K  5.0M   1% /run/lock
/dev/sda1                 511M  6.2M  505M   2% /boot/efi
tmpfs                     2.0G  304K  2.0G   1% /run/user/1000
```

Memory Usage

```
[redacted]@[redacted]-ThinkCentre-M710s:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:            19Gi       4.0Gi        12Gi       332Mi       3.2Gi        15Gi
Swap:          1.9Gi          0B       1.9Gi
```

Network Configuration

```
[redacted]@[redacted]-ThinkCentre-M710s:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback [redacted] brd [redacted]
    inet [redacted] scope host lo
       valid_lft forever preferred_lft forever
    inet6 [redacted] scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether [redacted]
    inet [redacted] brd [redacted] scope global dynamic noprefixroute enp1s0
       valid_lft 604478sec preferred_lft 604478sec
    inet6 [redacted] scope global temporary dynamic
       valid_lft 3591sec preferred_lft 3591sec
    inet6 [redacted] scope global dynamic mngtmpaddr noprefixroute
       valid_lft 3591sec preferred_lft 3591sec
    inet6 [redacted] scope global temporary dynamic
       valid_lft 291sec preferred_lft 291sec
    inet6 [redacted] scope global dynamic mngtmpaddr noprefixroute
       valid_lft 291sec preferred_lft 291sec
    inet6 [redacted] scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether [redacted] brd [redacted]
    inet [redacted] brd [redacted] scope global docker0
       valid_lft forever preferred_lft forever
```